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Week 4~ Week8 Content Review

- **Week 4: Hardware Interrupts (Coding)**
 - Reference Materials: Zynq Book Tutorials.pdf (Slides 44-115, also Lab 2 and Lab 4), The Zynq eBook_Slides.pdf (Slides 59-74)
 - Generic Interrupt Controller (GIC)
 - Difference Between shared peripheral interrupts and private peripheral interrupts
 - Interrupt Control Distributor
 - Implementation of hardware interrupts
 - Implementation of AXI Timer Interrupt
- **Week 5: Data Queue**
 - Reference Material: Introduction to FreeRTOS_Slides.pdf (125-196), also Lab3
 - Implementation of a queue
 - Reading, writing order
 - Multiple sources queue implementation
 - Mailbox
 - Mutex (Coding)
- **Week 6 Software Timer, AXI Timer (Lab 4)**
 - Software timer
 - One-shot timer, auto-reload timer (coding)
 - Software timer vs AXI timer
 - AXI Timer Interrupt
 - Concatenation of button interrupt and AXI Timer Interrupt
 - Implementation of AXI Timer Interrupt
- **Week 8 – Time-based Measurement:**
 - Time-based Measurement
 - Time-based measurement can help decide directions and speed of a rotary application.
 - Potential issues in time-based measurement and provide an example of it.
 - Range
 - the range of the 16-bit input capture timer decides all possible motor speed.
 - Accuracy: the rotational speed of the shaft cannot be calculated with such consistent accuracy across the full range.
 - As the speed of rotation increases, the difficulty of detecting small changes in speed increases.
 - Sampling Rate
 - Because it is dependent on encoder pulses to trigger input, our example system can measure the motor speed no faster than the encoder pulse rate.
 - As the motor speed decreases, the time between new speed estimates falls.

At 3,000 RPM, the sampling rate is 5 kHz; at 2 RPM, it falls to just 0.333Hz.

- Drift: When you have two timers operating from different crystal oscillators, you must assume that the values in the timers will diverge over time.

Timer Calculation

- Suppose that an application uses an embedded processor to control the speed of a DC motor.
 - This motor may rotate at speeds in the range 1 to 3,000 revolutions per minute (RPM).
 - If the encoder produces 100 pulses with each revolution of the shaft.
 - what range of the input capture timer?

Queue Scheduling, Fixed Priority

Five FreeRTOS tasks (T) are created with initial priorities (P): T1-P1, T2-P2, T3-P2, T4-P3 and T5-P4 and a single queue Q of 5 elements. An Idle task is created by default. The tick interval is 5 msec (not the default 10 msec) and time slice scheduling is used and. Scheduling starts at 0 msec.

All send and receive queue functions for the tasks have an indefinite wait time.

- a) Every time, T1-P1 when running for 5 msec sends one element to Q at the end of its task
- b) Everytime, T2-P2 when running for 10 msec receives two elements from Q at the end of its task
- c) Everytime T3-P2 when running for 5 msec receives one element from Q at the end of its task
- d) Everytime T4-P3 when running for 10 msec sends one element to Q at the end of its task
- e) Everytime T5-P4 sets the delay parameter of vTaskDelayUntil for 45 msec at the beginning of its task and runs for 5 msec

